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USED IN LIEU OF
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SUPERSEDING
MIL-M-0038510/20B(USAF)
15 October 1973

MILITARY SPECIFICATION

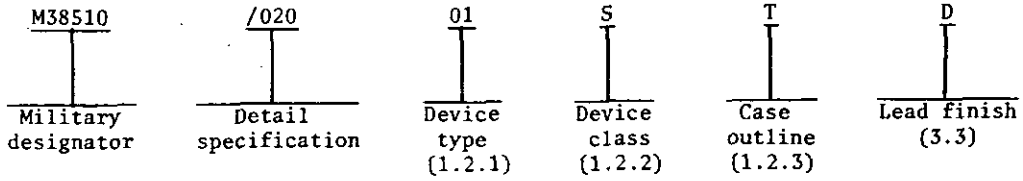
MICROCIRCUITS, DIGITAL, TTL, LOW POWER NAND GATES, MONOLITHIC SILICON

This limited coordination Military specification has been prepared by the Air Force Space and Missile Systems Organization based upon currently available technical information, but it has not been approved for promulgation as a coordinated revision of Military Specification MIL-M-38510/20A. It is subject to modification. However, pending its promulgation as a coordinated Military specification, it may be used in procurement.

1. SCOPE

1.1 Scope. This specification covers the requirements for monolithic silicon, TTL, low power, NAND gates, microcircuits for use in Air Force missile and space systems.

1.2 Part number. The complete part number shall be as shown in the following example:



1.2.1 Device type. The device type shall be as shown in the following:

Device type	Circuit	Similar commercial type
01	Single, 8-input positive NAND gate	54L30
02	Dual, 4-input positive NAND gates	54L20
03	Triple, 3-input positive NAND gates	54L10
04	Quadruple, 2-input positive NAND gates	54L00
05	Hex, 1-input inverter gates	54L04
06	Quadruple, 2-input positive NAND gates (open collector output)	54L01

1.2.2 Device class. The device class shall be class S as defined in MIL-M-0038510.

1.2.3 Case outline. The case outline shall be as specified in figure 10 of MIL-M-0038510 and shall be identified as outline T.

1.3 Absolute maximum ratings.

Supply voltage range	0 to 8.0 Vdc
Input voltage range	0 to 5.5 Vdc
Storage temperature range	-65°C to 150°C
Lead temperature (soldering, 10 seconds)	300°C
Ambient operating temperature range	-55°C to 125°C

1.4 Recommended operating conditions.

Ambient operating temperature range	-55°C to 125°C
Supply voltage	4.5 Vdc minimum to 5.5 Vdc maximum
Minimum high level input voltage	2.0 Vdc
Maximum low level input voltage	0.6 Vdc

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2. APPLICABLE DOCUMENTS

2.1 The following document, of the issue in effect on date of invitation for bids or request for proposal, forms a part of this specification to the extent specified herein.

SPECIFICATION

MILITARY

MIL-M-0038510 - Microcircuits, General Specification for.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

3. REQUIREMENTS

3.1 Detail specification. The individual item requirements shall be in accordance with MIL-M-0038510 and as specified herein.

3.2 Design, construction, weight and physical dimensions. The design, construction, weight and physical dimensions shall be as specified in MIL-M-0038510 and herein.

3.2.1 Logic diagram and terminal connections. The logic diagram and terminal conditions shall be as specified on figure 1.

3.2.2 Truth tables and logic equations. The truth tables and logic equations shall be as specified on figure 2.

3.2.3 Schematic circuit. The schematic circuit shall be as specified on figure 3.

3.2.4 Case outline. The case outline shall be as specified in 1.2.3.

3.2.5 Weight. The maximum weight shall be 1 gram.

3.3 Lead material and finish. The lead material and finish shall be in accordance with MIL-M-0038510.

3.4 Electrical performance characteristics. The electrical performance characteristics are as specified in tables I and II.

3.4.1 Delta drift limits. The delta drift limits shall be as specified in table I.

3.5 Screening. Screening shall be in accordance with table I of MIL-M-0038510 and as specified herein.

3.5.1 Interim electrical parameters. The interim electrical parameters shall consist of all the electrical characteristics tests as specified in table I. Each test shall be performed go, no-go at -55°C and 125°C. Each test shall also be read and recorded at 25°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2 Final electrical tests. The final electrical tests shall consist of all the electrical characteristics tests specified in table I. Each test shall be read and recorded at -55°C, 25°C and 125°C. For all tests, the devices shall meet the limits specified in table I.

3.5.2.1 Delta computation. Delta computations shall be performed for those parameters in table I which are marked with an asterisk. All devices shall meet the applicable delta limits specified in table I. (See section 6 in MIL-M-0038510.)

3.5.3 Switching characteristics tests. The switching characteristics tests specified in table II and figure 4 or 5 as applicable shall be performed at 25°C. All devices shall meet the limits specified in table II.

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TABLE I. (Device type 01) Electrical characteristics tests.

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	I_{IH1}		2.4 V	GND	5.5 V	GND	GND	GND	GND	GND	GND	GND					10	10X or 1 µA	µA
2*	↑		GND	2.4 V	↑	GND	GND	↓	↓	↓	↓	↑					↓	↓	↑
3*			↑	GND		2.4 V	GND	↓											
4*				↑		GND	2.4 V	GND	↓										
5*						↑	GND	2.4 V	GND	↓									
6*							↑	GND	2.4 V	GND	↓								
7*	↓		↓	↓	↑	↑	↑	↑	GND	2.4 V	GND	↑					↓	↓	↓
8*	I_{IH1}		GND	↓					↑	GND	2.4 V						10	10X or 1 µA	
9	I_{IH2}		5.5 V	GND		↓				↑	GND						100		
10	↑		GND	5.5 V		GND	↓				↑						↓		
11			↑	GND		5.5 V	GND	↓											
12				↑		GND	5.5 V	GND	↓										
13						↑	GND	5.5 V	GND	↓									
14							↑	GND	5.5 V	GND	↓								
15								GND	GND	5.5 V	GND								
16	I_{IH2}		GND	GND	5.5 V	GND	GND	GND	GND	GND	5.5 V						100		µA
17*	V_{OL}		2.0 V	2.0 V	4.5 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V	2.0 V		2.0 mA				0.3	±.03	V
18*	V_{OH}		0.7 V	5.5 V	↑	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		-.1 mA			2.4		±.12	↑
19*	↑		5.5 V	0.7 V		5.5 V	5.5 V	↓	↓	↓	↓		↓			↓		↓	
20*			↑	5.5 V		0.7 V	5.5 V	↓											
21*				↑		5.5 V	0.7 V	5.5 V	↓										
22*			↓	↓	↓	5.5 V	5.5 V	0.7 V	5.5 V	↓	↓		↓			↓		↓	
23*			5.5 V	5.5 V	4.5 V	5.5 V	5.5 V	5.5 V	0.7 V	5.5 V	5.5 V		.1 mA			2.4		±.12	V
24*			5.5 V	5.5 V	4.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.7 V	5.5 V		.1 mA			2.4		±.12	V
25*	V_{OH}		5.5 V	5.5 V	4.5 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.7 V		-.1 mA			2.4		±.12	V
26*	I_{IL}		0.3 V	5.5 V	5.5 V	5.5 V	↑	↑	↑	↑	5.5 V					-60	-180	±10	µA
27	I_{IL}		5.5 V	0.3 V	5.5 V	5.5 V	↓	↓	↓	↓	5.5 V					-60	-180	±10	µA
28	I_{IL}		5.5 V	5.5 V	5.5 V	0.3 V	5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	GND				-60	-180	±10	µA

See notes at end of device type 01.

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TABLE I. (Device type 01) Electrical characteristics tests. - Continued

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29	I_{IL}		5.5 V	5.5 V	5.5 V	5.5 V	0.3 V	5.5 V	5.5 V	5.5 V	5.5 V	GND				-60	-180	±10	µA
30	↕		↕	↕	↕	↕	5.5 V	0.3 V	5.5 V	5.5 V	5.5 V	↕				↕	↕	↕	↕
31							↕	5.5 V	0.3 V	5.5 V	5.5 V								
32								5.5 V	5.5 V	0.3 V	5.5 V								
33	I_{IL}		5.5 V	5.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V	0.3 V					-60	-180	±10	µA
34	I_{OS}		GND	GND		GND	GND	GND	GND	GND	GND		GND			-3	-15		mA
35	I_{CCL}		5.0 V	5.0 V	↕	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	5.0 V	↕					510		µA
36	I_{CCH}		GND	GND	5.5 V	GND	GND	GND	GND	GND	GND	GND					330		µA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. (Device type 02) Electrical characteristics tests.

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
1*	V_{OL}	2.0 V	2 mA		4.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	2.0 V	2.0 V	2.0 V		.3	±.03	V
2*	V_{OL}	5.5 V					2.0 V	2.0 V	2.0 V	2.0 V	2 mA		5.5 V	5.5 V	5.5 V		.3	±.03	
3*	V_{OH}	.7 V	-.1 mA				5.5 V	5.5 V	5.5 V	5.5 V			5.5 V	5.5 V		2.4		±.12	
4*		5.5 V											0.7 V	5.5 V					
5*													5.5 V	0.7 V	5.5 V				
6*			-.1 mA				5.5 V							5.5 V	0.7 V				
7*							0.7 V	5.5 V			-.1 mA				5.5 V				
8*							5.5 V	0.7 V	5.5 V										
9*							5.5 V	5.5 V	0.7 V	5.5 V									
10*	V_{OH}	5.5 V			4.5 V		5.5 V	5.5 V	5.5 V	0.7 V	-.1 mA		5.5 V	5.5 V	5.5 V	2.4		±.12	V
11	I_{OS}	GND	GND		5.5 V								GND	GND	GND	-3	-15		mA
12	I_{OS}						GND	GND	GND	GND	GND					-3	-15		mA
13*	I_{IH1}	2.4 V											GND	GND	GND		10	10X or 1 µA	µA
14*		GND											2.4 V	GND	GND				
15*													GND	2.4 V	GND				
16*							GND							GND	2.4 V				
17*							2.4 V	GND							GND				
18*							GND	2.4 V	GND										
19*								GND	2.4 V	GND									
20*	I_{IH1}	GND							GND	2.4 V							10	10X or 1 µA	
21	I_{IH2}	5.5 V								GND			GND				100		
22		GND											5.5 V	GND					
23													GND	5.5 V	GND				
24							GND							GND	5.5 V				
25							5.5 V	GND							GND				
26							GND	5.5 V	GND										
27							GND	GND	5.5 V	GND									
28	I_{IH2}	GND			5.5 V		GND	GND	GND	5.5 V		GND	GND	GND	GND		100		µA

See notes at end of device type 02.

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TABLE I. (Device type 02) Electrical characteristics tests. - Continued

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29	I_{IL}	0.3 V			5.5 V		5.5 V	5.5 V	5.5 V	5.5 V		GND	5.5 V	5.5 V	5.5 V	-60	-180		µA
30	↕	5.5 V			↕		↕	↕	↕	↕		↕	0.3 V	5.5 V	5.5 V	↕	↕		↕
31	↕	↕			↕		↓	↕	↕	↕		↕	5.5 V	0.3 V	5.5 V	↕	↕		↕
32	↓	↓			↓		5.5 V	↓	↓	↓		↓	5.5 V	5.5 V	0.3 V	↓	↓		↓
33	I_{IL}	5.5 V			5.5 V		0.3 V	5.5 V	5.5 V	5.5 V		GND	5.5 V	5.5 V	5.5 V	-60	-180		µA
34	I_{IL}	5.5 V			5.5 V		5.5 V	0.3 V	5.5 V	5.5 V		GND	5.5 V	5.5 V	5.5 V	-60	-180		µA
35	I_{IL}	5.5 V			↕		5.5 V	5.5 V	0.3 V	5.5 V		↕	5.5 V	5.5 V	5.5 V	-60	-180		µA
36	I_{IL}	5.5 V			↕		5.5 V	5.5 V	5.5 V	0.3 V		↕	5.5 V	5.5 V	5.5 V	-60	-180		µA
37	I_{CCL}	5.5 V			↕		5.5 V	5.5 V	5.5 V	5.5 V		↕	5.5 V	5.5 V	5.5 V		1.02		mA
38	I_{CCH}	GND			5.5 V		GND	GND	GND	GND		GND	GND	GND	GND		0.40		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. (Device type 03) Electrical characteristics tests.

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	I_{IH1}	2.4 V	GND		5.5 V		GND	GND	GND	GND	GND	GND	GND		GND		10	10X or 1 µA	µA
2*		GND	2.4 V				GND								GND				
3*			GND				GND								2.4 V				
4*							2.4 V	GND							GND				
5*								2.4 V	GND										
6*								GND	2.4 V	GND									
7*									GND	2.4 V	GND								
8*										GND	2.4 V		GND						
9*	I_{IH1}	GND									GND		2.4 V				10	10X or 1 µA	
10	I_{IH2}	5.5 V	GND										GND				100		
11		GND	5.5 V												GND				
12			GND				2.4 V								5.5 V				
13							5.5 V	GND							GND				
14							GND	5.5 V	GND										
15								GND	5.5 V	GND									
16									GND	5.5 V	GND								
17									GND	GND	5.5 V		GND						
18	I_{IH2}	GND	GND		5.5 V		GND	GND	GND	GND	GND		5.5 V		GND		100		µA
19*	V_{OL}	2.0 V	2.0 V	2 mA	4.5 V		5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5		2.0 V		0.3	±.03	V
20*	V_{OL}	5.5 V	5.5 V			2 mA	2.0 V	2.0 V	2.0 V	5.5 V	5.5 V		5.5 V		5.5 V		0.3	±.03	
21*	V_{OL}	5.5 V	5.5 V				5.5 V	5.5 V	5.5 V	2.0 V	2.0 V		2.0 V	2 mA			0.3	±.03	
22*	V_{OH}	0.7 V	5.5 V	-100 µA			5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V			2.4		±.12	
23*	V_{OH}	5.5 V	0.7 V	-100 µA			5.5 V	5.5 V	5.5 V	5.5 V	5.5 V				5.5 V				
24*	V_{OH}		5.5 V	-100 µA			5.5 V	5.5 V	5.5 V						0.7 V				
25*						-100 µA	0.7 V	5.5 V							5.5 V				
26*						-100 µA	5.5 V	0.7 V											
27*						-100 µA	5.5 V	5.5 V	0.7 V	5.5 V									
28*	V_{OH}	5.5 V	5.5 V		4.5 V		5.5 V	5.5 V	5.5 V	0.7 V	5.5 V	GND	5.5 V	-100 µA	5.5 V	2.4 V		±.12	V

See notes at end of device type 03.

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TABLE I. (Device type 03) Electrical characteristics tests. - Continued

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29*	V_{OH}	5.5 V	5.5 V		4.5 V		5.5 V	5.5 V	5.5 V	5.5 V	0.7 V	GND	5.5 V	-100 µA	5.5 V	2.4		±.12	V
30*	V_{OH}	5.5 V	5.5 V		4.5 V		↑	↑	↑	↑	5.5 V	↑	0.7 V	-100 µA	↓	2.4		±.12	V
31*	I_{IL}	0.3 V	5.5 V		5.5 V						↑		5.5 V		↓	-60	-180	±10	µA
32*	↑	5.5 V	0.3 V		↑		↓						↑		5.5 V	↑	↑	↑	↑
33*		↑	5.5 V				5.5 V	↓							0.3 V				
34*			↑				0.3 V	5.5 V	↓						5.5 V				
35*							5.5 V	0.3 V	5.5 V	↓					↓				
36*							↑	5.5 V	0.3 V	5.5 V	↑								
37*								↑	5.5 V	0.3 V	5.5 V		↑						
38*	↓								↑	5.5 V	0.3 V		5.5 V			↓	↓	↓	↓
39*	I_{IL}	↓	↓				↓	↓	↓	5.5 V	5.5 V		0.3 V		↓	-60	-180	±10	µA
40	I_{CCL}	5.5 V	5.5 V				5.5 V	5.5 V	5.5 V	5.5 V	5.5 V		5.5 V		5.5 V		1.53		mA
41	I_{CCH}	GND	GND				GND	GND	GND	GND	GND		GND		GND		0.60		↑
42	I_{OS}	GND	GND	GND											GND	-3	-15		
43	I_{OS}				↓	GND	GND	GND	GND			↓				-3	-15		↓
44	I_{OS}				5.5 V					GND	GND	GND	GND	GND		-3	-15		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. (Device type 04) Electrical characteristics tests.

1/2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
1*	V_{OH}	.7 V	5.5 V	-100 µA	4.5 V		5.5 V	5.5 V		5.5 V	5.5 V	GND	5.5 V	5.5 V		2.4		±.12	V
2		5.5 V	.7 V	-100 µA			5.5 V	5.5 V											
3*			5.5 V			-100 µA	.7 V	5.5 V										±.12	
4						-100 µA	5.5 V	.7 V		5.5 V									
5*								5.5 V	-100 µA	.7 V	5.5 V							±.12	
6									-100 µA	5.5 V	.7 V		5.5 V						
7*											5.5 V		.7 V	5.5 V	-100 µA			±.12	
8	V_{OH}	5.5 V	5.5 V										5.5 V	.7 V	-100 µA	2.4			
9*	V_{OL}	2.0 V	2.0 V	2 mA			5.5 V	5.5 V						5.5 V			.3	±.03	
10*		5.5 V	5.5 V			2 mA	2.0 V	2.0 V		5.5 V	5.5 V			5.5 V					
11*		5.5 V	5.5 V				5.5 V	5.5 V	2 mA	2.0 V	2.0 V		5.5 V	5.5 V					
12*	V_{OL}	5.5 V	5.5 V		4.5 V					5.5 V	5.5 V		2.0 V	2.0 V	2 mA		.3	±.03	V
13*	I_{IL}	.3 V	5.5 V		5.5 V								5.5 V	5.5 V		-60	-180	±.10	µA
14*		5.5 V	.3 V				5.5 V												
15*			5.5 V				.3 V	5.5 V											
16*							5.5 V	.3 V		5.5 V	5.5 V								
17*								5.5 V		.3 V	5.5 V								
18*										5.5 V	.3 V		5.5 V	5.5 V					
19*										5.5 V	5.5 V		.3 V	5.5 V					
20*	I_{IL}	5.5 V	5.5 V				5.5 V	5.5 V		5.5 V	5.5 V		5.5 V	.3 V		-60	-180	±10	
21*	I_{IH1}	2.4 V	GND				GND	GND		GND	GND		GND	GND			10	10X or 1 µA	
22*		GND	2.4 V				GND	GND		GND	GND		GND	GND					
23*		GND	GND				2.4 V	GND		GND	GND		GND						
24*		GND	GND				GND	2.4 V		GND	GND		GND						
25*		GND	GND				GND	GND		2.4 V	GND		GND						
26*		GND	GND				GND	GND		GND	2.4 V		GND						
27*	I_{IH1}	GND	GND		5.5 V		GND	GND		GND	GND	GND	2.4 V	GND			10	10X or 1 µA	µA

See notes at end of device type 04.

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TABLE I. (Device type 04) Electrical characteristics tests. - Continued

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
28*	I_{IH1}	GND	GND		5.5 V		GND	GND		GND	GND	GND	GND	2.4 V			10	10X or 1 µA	µA
29	I_{IH2}	5.5 V	GND		↑		GND	GND		GND	GND	↑	GND	GND			100		↕
30	↕	GND	5.5 V		↕		GND	GND		GND	GND	↕	GND	↑			↕		↕
31	↕	GND	GND		↕		5.5 V	GND		GND	GND	↕	GND	↕			↕		↕
32	↕	GND	GND		↕		GND	5.5 V		GND	GND	↕	GND	↓			↕		↕
33	↕	GND	GND		↕		GND	GND		5.5 V	GND	↕	GND	GND			↕		↕
34	↕	GND	GND		↕		GND	GND		GND	5.5 V	↕	GND	GND			↕		↕
35	↓	GND	GND		↕		GND	GND		GND	GND	↕	5.5 V	GND			↓		↕
36	I_{IH2}	GND	GND		↕		GND	GND		GND	GND	↕	GND	5.5 V			100		µA
37	I_{OS}	GND	GND	GND	↕		5.5 V	5.5 V		5.5 V	5.5 V	↕				-3	-15		mA
38	↑	GND	GND		↕	GND	GND	GND		5.5 V	5.5 V	↕					↑		↕
39	↓	GND	GND		↕				GND	GND	GND	↕					↓		↕
40	I_{OS}	GND	GND		↕							↕	GND	GND	GND		-15		↕
41	I_{CCL}	5.5 V	5.5 V		↓		5.5 V	5.5 V		5.5 V	5.5 V	↕	5.5 V	5.5 V			2.04		↕
42	I_{CCH}	GND	GND		5.5 V		GND	GND		GND	GND	GND	GND	GND			0.80		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. (Device type 05) Electrical characteristics tests.

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS 1	2	3	4	5	6	7	8	9	10	11	12	13	14	3/ LIMITS MIN	MAX	1/ DELTA Δ	UNITS
1*	V_{OL}	2.0 V		5.5 V	4.5 V	5.5 V		5.5 V		5.5 V		GND		5.5 V	2 mA		0.3	±.03	V
2*	↑	5.5 V	2 mA	2.0 V	↑	5.5 V		5.5 V		5.5 V		↑		↑			↑	↑	↑
3*		↑		5.5 V		2.0 V	2 mA	5.5 V		5.5 V									
4*				↑		5.5 V		2.0 V	2 mA	5.5 V									
5*						↑		5.5 V		2.0 V	2 mA			5.5 V					
6*	V_{OL}	5.5 V						↑		5.5 V			2 mA	2.0 V			0.3	±.03	
7*	V_{OH}	0.7 V		5.5 V		↓				↑				5.5 V	-.1 mA	2.4		±.120	
8*	↑	5.5 V	-.1 mA	0.7 V		5.5 V		↓						↑		↑		↑	
9*		↑		5.5 V		0.7 V	-.1 mA	5.5 V		↓									
10*				↑		5.5 V		0.7 V	-.1 mA	5.5 V				↑					
11*	↓	↓		↓	↓	5.5 V		5.5 V		0.7 V	-.1 mA			5.5 V					
12*	V_{OH}	5.5 V		5.5 V	4.5 V	5.5 V		5.5 V		5.5 V			-.1 mA	0.7 V		2.4		±.120	V
13	I_{OS}	GND			5.5 V										GND	-3	-15		mA
14	↑		GND	GND	↑											↑	↑		↑
15						GND	GND												
16								GND	GND										
17	↓									GND	GND								
18	I_{OS}												GND	GND		-3	-15		mA
19*	I_{IH1}	2.4		GND		GND		GND		GND				GND			10	10X or 1 µA	µA
20*	↑	GND		2.4 V		GND		GND		GND				GND			10	10X or 1 µA	↑
21*		↑		GND		2.4 V		GND		GND				↑				↑	
22*				↑		GND		2.4 V		GND									
23*						↑		GND		2.4 V				GND					
24*	I_{IH1}	GND						↑		GND				2.4 V			10	10X or 1 µA	
25	I_{IH2}	5.5 V		GND		↓				↑				GND			100		
26	↑	GND		5.5 V		GND		↓						GND			↑		
27		GND		GND	↓	5.5 V		GND		↓				GND					
28	I_{IH2}	GND		GND	5.5 V	GND		5.5 V		GND		GND		GND			100		µA

See notes at end of device type 05.

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TABLE I. (Device type 05) Electrical characteristics tests. - Continued

1/ 2/ TEST NUMBER	TEST NAME	PIN NUMBERS														3/ LIMITS		1/ DELTA Δ	UNITS
		1	2	3	4	5	6	7	8	9	10	11	12	13	14	MIN	MAX		
29	I_{IH2}	GND		GND	5.5 V	GND		GND		5.5 V		GND		GND			100		μA
30	I_{IH2}	GND		GND	↑	GND		GND		GND		↑		5.5 V			100		↑
31	I_{IL}	0.3 V		5.5 V		5.5 V		5.5 V		5.5 V				5.5 V		-60	-180		
32	↑	5.5 V		0.3 V		5.5 V		5.5 V		5.5 V				5.5 V		-60	-180		
33		↑		5.5 V		0.3 V		5.5 V		5.5 V				5.5 V		-60	-180		
34				5.5 V		5.5 V		0.3 V		5.5 V				5.5 V		-60	-180		
35	↓	↓		5.5 V		5.5 V		5.5 V		0.3 V				5.5 V		-60	-180		↓
36	I_{IL}	5.5 V		5.5 V		5.5 V		5.5 V		5.5 V				0.3 V		-60	-180		μA
37	I_{CCL}	5.5 V		5.5 V	↓	5.5 V		5.5 V		5.5 V		↓		5.5 V			3.10		mA
38	I_{CCH}	GND		GND	5.5 V	GND		GND		GND		GND		GND			1.20		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE I. (Device type 06) Electrical characteristics tests.

Test No. 1/2/	Characteristics	Symbol	Conditions	Limits 3/		Delta (Δ) 1/	Units
				Min	Max		
1	High-level input voltage	V_{IH}	-	2			V
2	Low-level input voltage	V_{IL}	-		0.6		V
3*	High-level output current	I_{OH}	V_{CC} = 4.5 V, V_{IL} = 0.6 V, V_{OH} = 5.5 V		50	10X or 5 μA	μA
4*	Low-level output voltage	V_{OL}	V_{CC} = 4.5 V, V_{IH} = 2 V, I_{OL} = 2 mA		0.3	0.03	V
5*	High-level input current (each input)	I_{IH1}	V_{CC} = 5.5 V, V_{IN} = 2.4 V		10	10X or 1 μA	μA
6	High-level input current (each input)	I_{IH2}	V_{CC} = 5.5 V, V_{IN} = 5.5 V		100		μA
7	Low-level input current (each input)	I_{IL}	V_{CC} = 5.5 V, V_{IN} = 0.3 V	-.06	-.18		mA
8	Supply current, high-level output	I_{CCH}	V_{CC} = 5.5 V V_{IN} = 0		0.8		mA
9	Supply current, low-level output	I_{CCL}	V_{CC} = 5.5 V, V_{IN} = 5 V		2.04		mA

1/ Delta computation shall be performed on parameters marked with an asterisk (*) with data read and recorded ("10X" denotes "10 times the initial measurement"). Where two delta values are indicated for one parameter, the greater value shall apply.
2/ Test sequence is optional with the manufacturer.
3/ Unless otherwise specified, the limits apply at T_A = -55°C to +125°C.

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TABLE II. (Device type 01) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	90	ns
				125°C, -55°C	3	140	
Propagation delay time low to high level output	t_{PLH}			25°C	3	60	ns
				125°C, -55°C	3	90	

TABLE II. (Device type 02) Switching characteristics.

Characteristic	Symbol	Conditions 1/	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	60	ns
				125°C, -55°C	3	90	
Propagation delay time low to high level output	t_{PLH}			25°C	3	60	ns
				125°C, -55°C	3	90	

1/ Test each gate.

TABLE II. (Device type 03) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	60	ns
				125°C, -55°C	3	90	
Propagation delay time low to high level output	t_{PLH}			25°C	3	60	ns
				125°C, -55°C	3	90	

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TABLE II. (Device type 04) Switching characteristics.

Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	V_{CC} = 5.0 V C_L = 50 pF R_L = 4 kΩ ±5%	4	25°C	3	60	ns
				125°C, -55°C	3	90	
Propagation delay time low to high level output	t_{PLH}			25°C	3	60	ns
				125°C, -55°C	3	90	

TABLE II. (Device type 05) Switching characteristics.

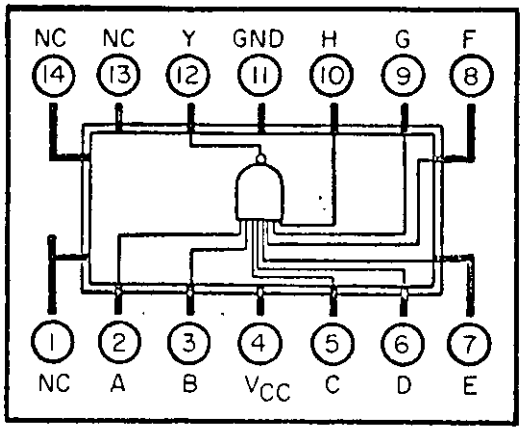
Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time high to low level output	t_{PHL}	C_L = 50 pF R_L = 4 kΩ ±5% V_{CC} = 5.0 V	4	25°C	3	60	ns
				125°C, -55°C	3	90	
Propagation delay time low to high level output	t_{PLH}			25°C	3	60	ns
				125°C, -55°C	3	90	

TABLE II. (Device type 06) Switching characteristics.

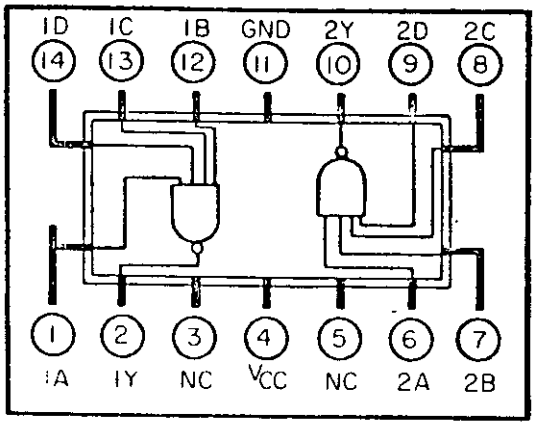
Characteristic	Symbol	Conditions	Test figure	Temperature	Limits		Units
					Min	Max	
Propagation delay time low to high level output	t_{PLH}	C_L = 50 pF R_L = 4 kΩ ±5% V_{CC} = 5.0 V	5	25°C	3	90	ns
				125°C, -55°C	3	140	
Propagation delay time high to low level output	t_{PHL}			25°C	3	66	ns
				125°C, -55°C	3	99	

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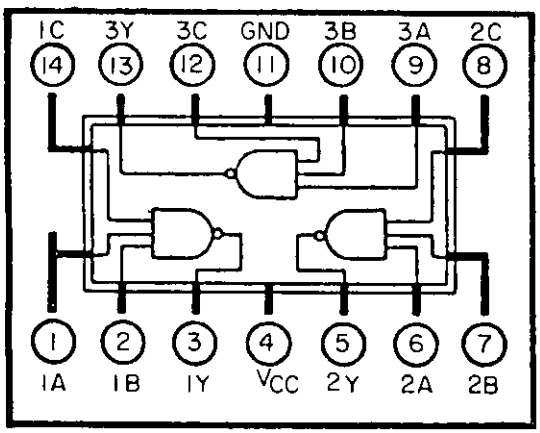
Device type 01



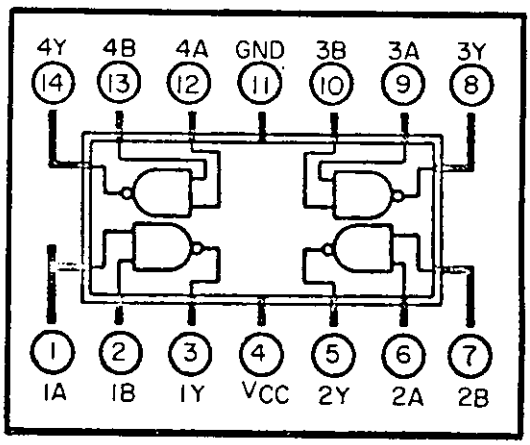
Device type 02



Device type 03



Device types 04 and 06



Device type 05

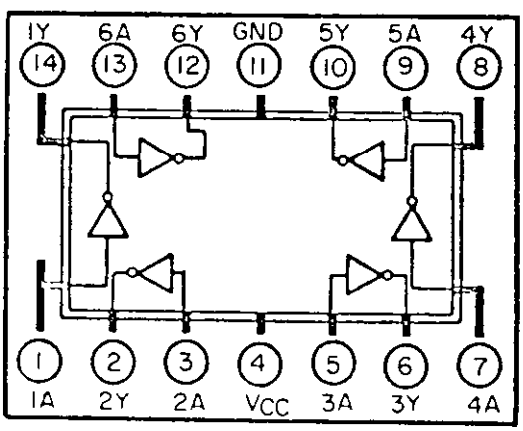


FIGURE 1. Logic diagrams and terminal connections.

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Device type 01

Truth table								
Input								Output
A	B	C	D	E	F	G	H	Y
H	H	H	H	H	H	H	H	L
All other combinations of H and L at the inputs give H output.								

Positive logic $Y = \overline{ABCDEFGH}$

Device type 05

Truth table each gate	
Input	Output
A	Y
L	H
H	L

Positive logic $Y = \overline{A}$

Device type 03

Truth table			
Input			Output
A	B	C	Y
L	L	L	H
H	L	L	H
L	H	L	H
H	H	L	H
L	L	H	H
H	L	H	H
L	H	H	H
H	H	H	L

Positive logic $Y = \overline{ABC}$

Device type 02

Truth table				
Input				Output
A	B	C	D	Y
L	L	L	L	H
H	L	L	L	H
L	H	L	L	H
H	H	L	L	H
L	L	H	L	H
H	L	H	L	H
L	H	H	L	H
H	H	H	L	H
L	L	L	H	H
H	L	L	H	H
L	H	L	H	H
H	H	L	H	H
L	L	H	H	H
H	L	H	H	H
L	H	H	H	H
H	H	H	H	L

Positive logic $Y = \overline{ABCD}$

Device types 04 and 06

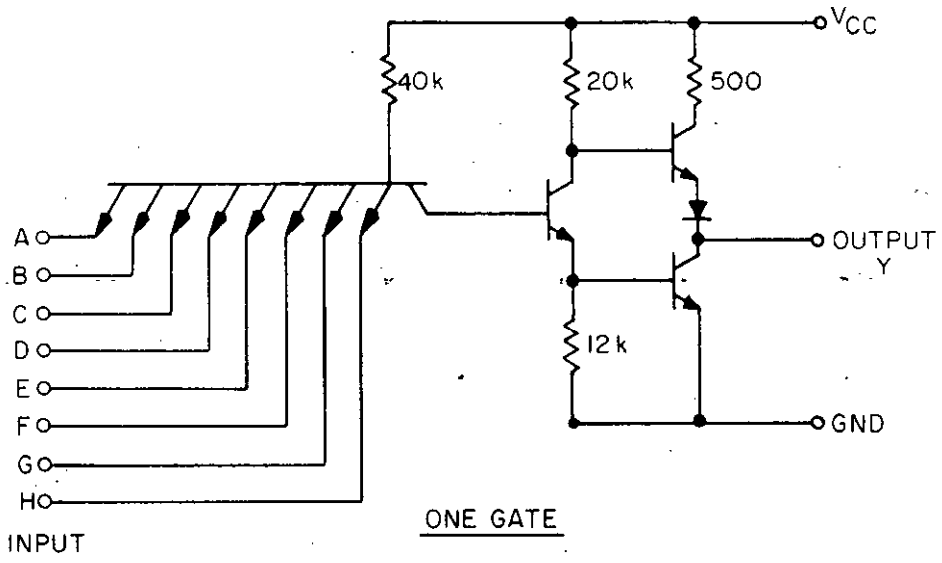
Truth table each gate		
Input		Output
A	B	Y
L	L	H
H	L	H
L	H	H
H	H	L

Positive logic $Y = \overline{AB}$

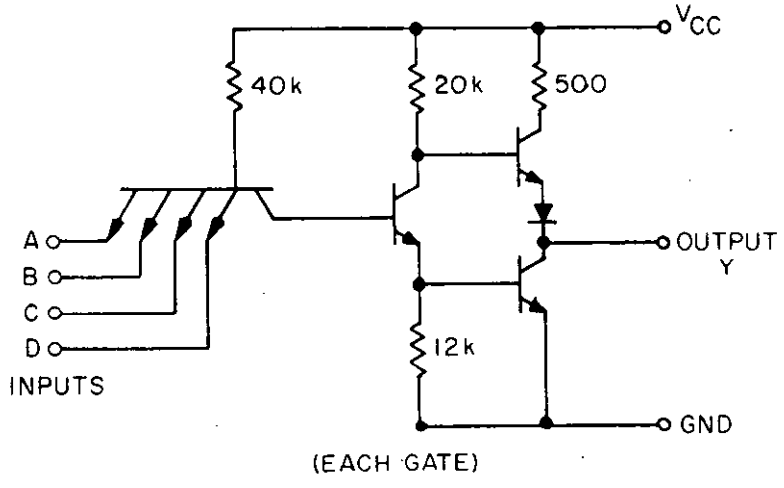
FIGURE 2. Truth tables and logic equations.

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Device type 01



Device type 02

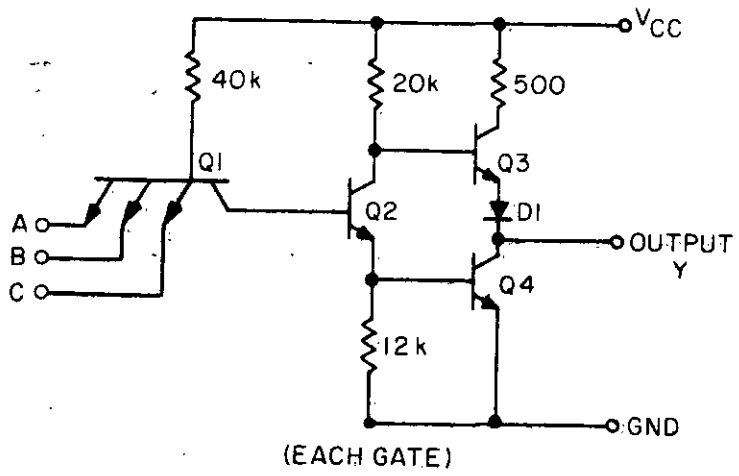


NOTE: All resistor values nominal and expressed in ohms, k = 1000.

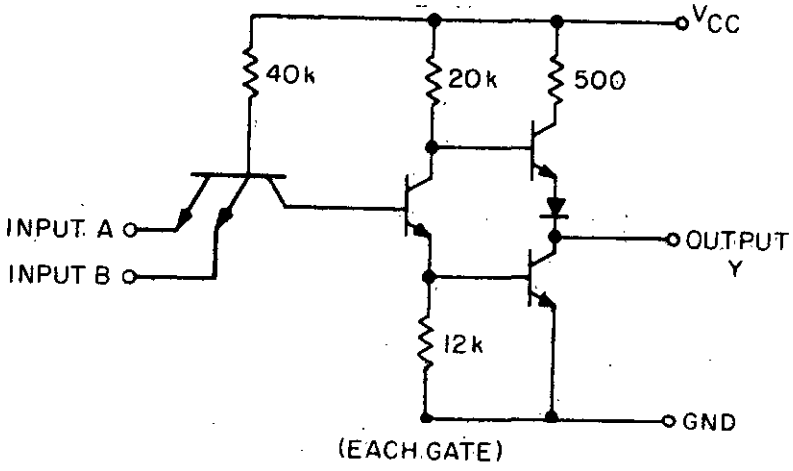
FIGURE 3. Schematic circuit.

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Device type 03



Device type 04

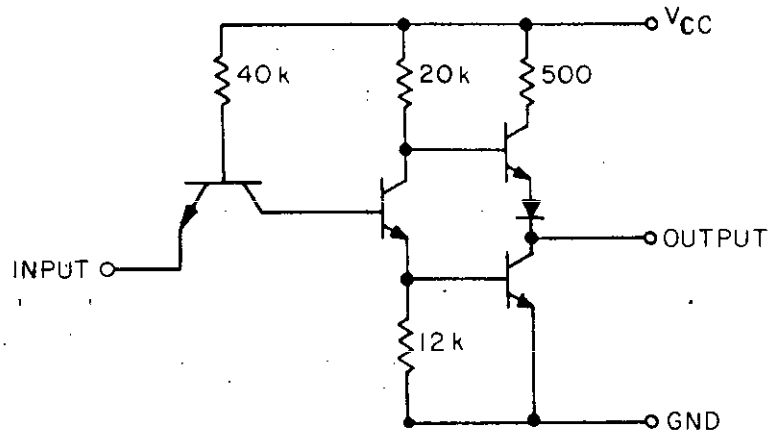


NOTES:
1. Component values shown are nominal
2. All resistance values in ohms, k = 1000 ohms.

FIGURE 3. Schematic circuit - Continued.

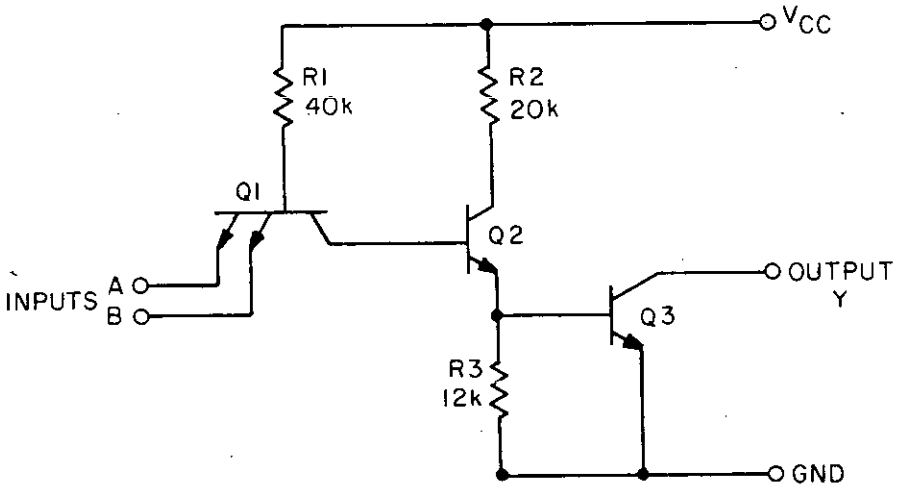
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Device type 05



NOTES:
1. One gate of six
2. All resistor values nominal and expressed in ohms, k = 1000 ohms.

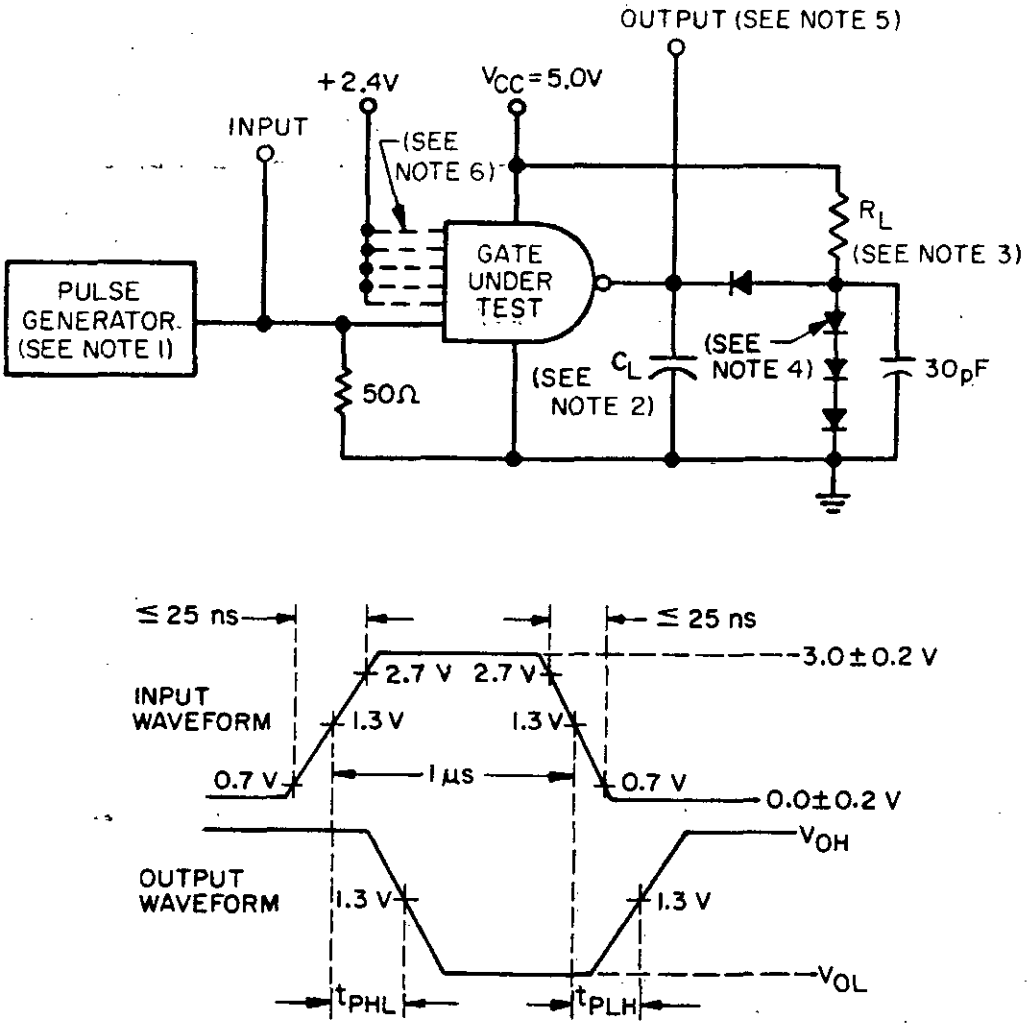
Device type 06



NOTES:
1. All resistor values are nominal and in ohms, k = 1000 ohms.
2. Only one gate is shown.

FIGURE 3. Schematic circuit - Continued.

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Check the source to verify that this is the current version before use.

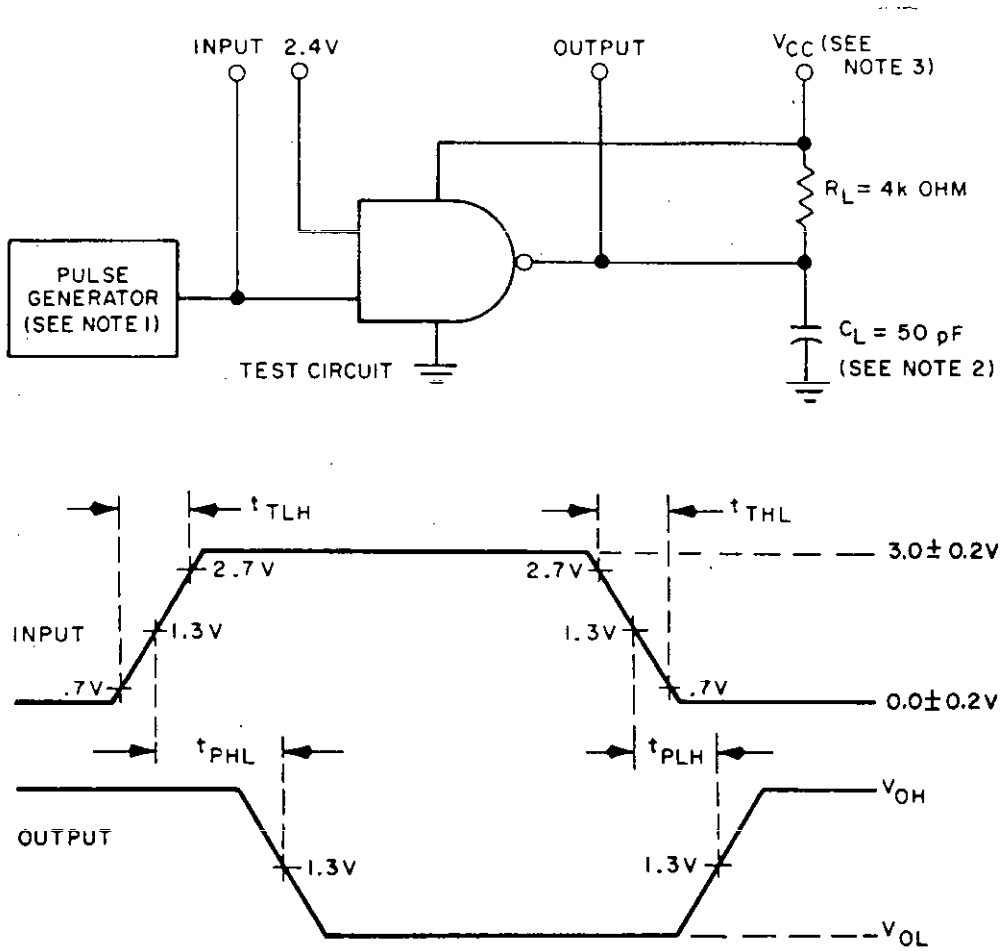


Device types 01, 02, and 05

NOTES:
1. The pulse generator has the following characteristics: V_{IH} = 3 V ±.2 V, V_{IL} = 0 V ±.2 V, t_p = 1 µs, PRR = 1 MHz, Z_{out} = 50Ω, and t_{THL} = $t_{TLH} \leq 60$ ns.
2. C_L = 50 pF minimum, including scope probe, wiring, and stray capacitance without package in test fixture.
3. R_L = 4 kΩ ±5%.
4. All diodes are 1N3064 or equivalent.
5. Voltage measurements are to be made with respect to network ground terminal.
6. The input signal is applied to one input and all other inputs are at 2.4 V.

FIGURE 4. Switching time test circuit and waveform (device types 01, 02, 03, 04 and 05.

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Device type 06

NOTES:
1. The pulse generator has the following characteristics: 3 V ±.2 V, V_{IL} = 0 V, $t_{TLH} = t_{THL} \leq 25$ ns, PRR = 1 MHz and $Z_{out} \approx 50\Omega$.
2. C_L = 50 PF minimum including scope probe, wiring, and stray capacitance, without package in fixture.
3. V_{CC} = 5 V and T_A = 25 ~ C.

FIGURE 5. Switching time test circuit and waveform (device type 06).

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3.6 Extended life tests. Extended life tests shall be performed in accordance with MIL-M-0038510.

3.7 Marking. Marking shall be in accordance with MIL-M-0038510 and 1.2 herein.

4. PRODUCT ASSURANCE PROVISIONS

4.1 Product assurance provisions. Product assurance shall be in accordance with MIL-M-0038510.

5. PREPARATION FOR DELIVERY

5.1 Preparation for delivery. Preparation for delivery shall be as specified in MIL-M-0038510.

6. NOTES

6.1 Notes. Notes are as specified in MIL-M-0038510.

Custodian:
Air Force - 19

Review activities:
Air Force - 17, 19
DSA - ES

Preparing activity:
Air Force - 19

Agent:
DSA - ES

(Project 5962-F088-25)

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